

Sathvika Allenki

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EDUCATION

The University of Texas at Dallas

MS in Computer Science, specialization in Data Science

B V Raju Institute of Technology, Narsapur, Hyderabad

Bachelor of Technology, Computer Science and Engineering

Aug 2023 – May 2025

GPA 3.71

Jul 2017 – Jun 2021

GPA 3.81

PROFESSIONAL EXPERIENCE

DataReveal AI | Plano, TX, USA

Feb 2025 – May 2025

AI Engineer Intern

- Developed Gen AI-based solutions for HR data analysis using Python and LLMs, improving data processing efficiency in the **Precision HR project**.
- Collaborated with cross-functional teams to design and build scalable **PySpark and SQL** data pipelines. Integrated computer vision solutions and optimized model performance based on testing feedback, enhancing decision-making and streamlining complex workflows.

The University of Texas at Dallas | Richardson, TX, USA

Jan 2024 – Feb 2025

Computer Science Teaching Assistant

- Served as a Teaching Assistant for CS 1436: Programming Fundamentals in C++ and CS 3354: Software Engineering courses.
- Conducted programming language sessions on C, C++, Python, Java, and Machine Learning for 150 students.
- Led the Deep Dive AI Summer Workshop at UT Dallas, teaching core AI concepts and hands-on projects.

Micron Technology | Hyderabad, India

Jul 2021 – Jul 2023

Software Engineer

- Successfully upgraded Teamcenter from **version 12.0 to 13.2**, ensuring seamless data migration and compatibility with 0% data loss. Created virtual machines, set up sandboxes, and performed daily deployments to maintain a stable and efficient development environment.
- Enhanced codebase efficiency by refactoring legacy **batch scripts to PowerShell**, adhering to modern design principles, and reducing the number of code files by 60%.
- Spearheaded a **CI/CD project at Micron**, customizing deployment processes. This initiative reduced deployment time by 30%, enhanced overall system stability, and reduced **post-deployment issues by 20%**.
- Integrated AppDynamics to monitor server performance in the pipeline environment, ensuring optimal system health and performance of critical software systems, resulting in 100% uptime.

Smart Interviews | Hyderabad, India

Apr 2021 – Jun 2021

Student Mentor

- Conducted sessions to address DSA queries using **C++, Python** and **Java**, leading to an improvement in student performance.

TECHNICAL SKILLS

Languages/Scripting: C, C++, Java, Python, HTML, CSS, JavaScript, TypeScript, PHP, SQL, XML, Batch scripting, PowerShell

Frameworks/Libraries: React, Angular, Next.js, FastAPI, GraphQL, Pandas, NumPy, PySpark, TensorFlow, Matplotlib, NLTK, RAG

Tools & Platforms: Git, Jira, Android Studio, VS Code, Postman, AWS, SQL Workbench, VMware, Kafka, Teamcenter, Hadoop

PROJECTS

RAG-based Document Q&A System: RAG, Streamlit, LangChain, OpenAI

- Developed an interactive document reader using Streamlit and LangChain with RAG architecture.
- Integrated Hugging Face embeddings and OpenAI's GPT-4 via LangChain Hub to enable intelligent PDF question-answering. Engineered a responsive UI to visualize extracted text and conversational responses in real-time

Recommendation System for E-commerce: Python, Collaborative Filtering, Scikit-learn

- Built a recommendation system using Python and Scikit-learn to suggest products based on user behavior.
- Combined collaborative filtering (ALS, cosine similarity) with content-based filtering (TF-IDF) for better accuracy. Preprocessed user-item data and validated model performance using RMSE.

Comparative Study on Deadlocks in Distributed Database Systems

- Implemented various deadlock detection and avoidance algorithms in a distributed relational database system and conducted a comparative study based on metrics such as detection accuracy, number of messages transmitted, delay, and time taken to detect and fix deadlocks.

Real-Time Log Analytics System

- Built an end-to-end real-time log analytics pipeline on Windows using Kafka, PySpark, Hadoop, and Hive to ingest, process, and analyze simulated server logs at scale.
- Implemented structured streaming with PySpark to parse and filter log levels (ERROR, WARN), storing raw and processed data in HDFS and querying insights via Hive on local infrastructure.